

Q1. Differentiate between ethics, law and professional responsibility with examples from the computing field.

Ans:

Aspect	Ethics	Law	Professional Responsibilities
Definition	Moral principles that guide what is right and wrong behaviour.	Rules and regulations created and enforced by the govt.	Duties & standards that professionals in a specific field are expected to follow.
Nature	Moral & value based	Legal and compulsory	Profession-specific and standard based
Authority for Enforcement	Society, personal conscience, public opinion.	Courts, police, govt. agencies.	Employers and professional organizations such as ACM & IEEE.
Consequences of violation	Loss of trust, damaged reputation	Fines, imprisonment, legal penalties	Job termination, suspension, loss of professional membership
Example in computing	Not misusing user data when technically possible	Hacking a system without permission is illegal.	Releasing properly tested and secure software to protect users,

Q2. Discuss any two historical cases of ethical failures in technology and explain the lessons learned from them.

Ans:

① Facebook - Cambridge Analytica Data Scandal (2018)

In 2018, it is revealed that Cambridge Analytica collected personal data from millions of fb users without proper consent. The data was obtained through a personality quiz app and later used for political advertising and voter profiling, including during the 2016 U.S. presidential election.

Ethical failures:

- Violation of user privacy
- Lack of informed consent
- Misuse of personal data
- Weak oversight of third party developers.

Lessons Learned :

- user data must be protected
- companies must ensure informed consent and transparency
- Strong data governance are essential
- Ethical responsibility goes beyond legal compliance

② Volkswagen Emissions Scandal (2015)

- In 2015, Volkswagen was found to have installed software in diesel vehicles that could detect when emissions testing was taking place. The software alternate engine performance during tests to meet environmental standards, while in normal ^{driving} conditions, the vehicles emitted pollutants far above legal limits

Ethical failures :

- Deliberate software manipulation,
- Deception of regulators and customers.
- prioritizing profit over environmental responsibility.

Lessons Learned:

- Technology must not be used to deceive.
- Honesty and accountability are essential in software development.
- Short term profit should not override ethics.

Q3. Explain the principle of software engg. Ethics, highlight the issues. Discuss how ACM/IEEE code of Ethics guides ethical decision making in soft. ware engg. practises.

Ans:

Principles of software engg. ethics:

- Software engg. ethics focuses on ensuring that software professionals acts responsibly toward society, clients, employers and users. The key principles include:

i) Software engineers must prioritize public safety, security & welfare.

ii) They must act in the best interest of clients and employers while maintaining honesty.

iii) Software should be reliable, safe and properly tested.

iv) Engineers must be truthful about system capabilities & limitations.

v) Sensitive information must be protected.

Issues related to professional responsibility:

- Maintaining confidentiality of user & company data.
- Avoiding conflict of interest
- Ensuring software safety and security.
- Reporting risks of errors instead of hiding them.
- Continuous learning to keep skills updated.

Role of ACM/IEEE code of ethics:

The ACM and IEEE provides a code of ethics that guides software engineers in ethical decision making.

The code emphasises:

- Acting in the public interest
- Avoiding harm
- Being honest and trustworthy
- Respecting privacy and intellectual property
- Improving professional competence

This code help engg. make ethical decision when facing dilemmas such^{as} whether to release incomplete software or how to handle sensitive data.

Q4. Differentiate Utilitarianism and Deontology. How would each approach handle a software privacy dilemma?

Ans:

Aspect	Utilitarianism	Deontology
Founders	Jeremy Bentham, John Stuart Mill	Immanuel Kant
Core idea	actions are right if they produce the greatest happiness for the greatest number.	actions are right if they follow moral duties and rules, regardless of consequences.
Focus	Consequences (outcome based)	Duties & moral principles (rule based)
Decision basis	maximize overall benefits, minimize harm.	Follow universal moral rules and respect human dignity.
flexibility	flexible; rules can change depending on results.	strict; some actions are always wrong (e.g., lying).

□ Software privacy Dilemma example.

Scenario: A company can share user data with researchers to improve public health, but users did not give explicit consent.

□ Utilitarian approach:

• Would evaluate overall benefits & harms.

• If sharing data helps millions of people and causes min. harm, it may consider sharing acceptable.

• Focus: Does this action create the greatest overall good?

□ Deontological (Kantian) approach:

• Would focus on moral duty & user rights.

• Sharing data without consent violates the duty to respect privacy & autonomy.

• Even if the outcome is beneficial, it is morally wrong because it treats users as a means to an end.

05. Explain Virtue ethics and describe how it can guide ethical decision making for software engineers.

Ans:

Virtue Ethics is an ethical framework that focuses on the character and moral qualities of a person rather than just rules or consequences.

It was mainly developed by Aristotle.

Instead of asking:

- Is this action legal? (Law)
- Will this action produce good results? (Utilitarianism)

Virtue Ethics asks:

- What would a good, honest and responsible person do?

It emphasizes developing good character virtues such

as:

- Honesty
- Integrity
- Responsibility
- Fairness
- Courage
- Compassion.

Virtue ethics guides software engg. by encouraging them to build strong moral character and professional integrity.

D) Honesty:

Be truthful about software limitations and risks.

ii) Responsibility:

Ensure software is safe and reliable before release.

iii) Integrity:

Refuse to participate in unethical projects.

iv) Courage:

Speak up if mgmt pressures you to release unsafe software.

v) Fairness:

Avoid bias in AI systems and algorithms.

Q7. What are the key principles of the ACM code of Ethics?

Why are such professional codes important for computing professionals?

Solⁿ:

The ACM code of Ethics provides guidelines for ethical behaviour in computing. Its key principle includes:

01. Contribute to society and human well-being

- computing professionals should ensure their work benefits society and avoids harm.

02. Avoid harm

- Prevent negative consequences such as security breaches, privacy violation or unsafe system.

03. Be honest & trustworthy

- Provide accurate info about system capabilities & limitations.

04. Respect privacy

- protect personal data and maintain confidentiality.

05. Accept Professional Responsibility

- Be accountable for one's work & its impact.

Professional codes like the ACM code of ethics are important because:

01. Provide Ethical Guidance: Help professionals make correct decisions in difficult situations.

02. Protect the Public: Ensure technology is developed safely & responsibly.

03. Promote Trust: Build confidence between professionals, clients & users.

04. Encourage Accountability: Remind professionals that they are responsible for the consequences of their

05. Maintain Professional Standards: Ensure quality, integrity, fairness in computing practices.

08. What is whistleblowing? Discuss the ethical considerations and challenges a computing professional might face when deciding to blow the whistle.

Defn: Whistleblowing is the act of reporting illegal, unethical or harmful activities with an organization to authorities, management, or the public.

Ethical considerations when:

01. Public Safety is at Risk: For ex, releasing unsafe soft. that may harm users.
02. Violation of Law or Ethics: Such as illegal data collection or security manipulation.
03. Professional Responsibility: Ethical codes (like ACM) encourage professionals to avoid harm and act in the public interest.
04. Duty of Loyalty vs Public Interest: Balancing loyalty to the employer with responsibility to society.

Challenges faced by computing Professionals :

01. Risk of Job loss : The employer may be fined

02. Legal and Financial Risks : Possible lawsuits or legal complications.

03. Workplace Retaliation : Harassment or isolation by colleagues.

04. Emotional Stress : Fear, anxiety and career uncertainty

10. Explain the differences between copyright, patents and trademarks in the context of software & digital products.

Aspects	Copyright	Patent	Trademark
i) Protects	Source code, design, content	New invention or technical method	Name, logo, brand
ii) Focus	Expression of idea	Functional innovation	Brand identity
iii) Need Registration	No (auto)	Yes	Yes (for strong protection)
iv) Duration	Long term (Life of author + many years)	About 20 yrs	Renewable (can last long)
v) Example	Copying app code	Patented Algorithm	App name / Logo

11. Compare Open Source Licensing with Proprietary Software Licensing. What are the ethical implications of each model.

Aspect	Open source Licensing	Proprietary Licensing
01. Access to source code	Source code is publicly available	Source code is closed and not accessible.
02. Modification	User can modify & improve the software	Modification is restricted & prohibited.
03. Distribution	Free redistribution is usually allowed	Redistribution is limited or forbidden
04. Cost	Often free or low cost.	Usually paid for subscription or buy
05. Example	Linux	Ms Windows

Ethical implications

Open source Model :

1. Promotes transparency & trust.
2. Improves security through public review
3. Supports equal access to technology

Challenges:

- Lack of clear accountability
- Sustainability and funding issues.

Proprietary Model :

1. Clear ownership and accountability
2. Strong quality control
3. Encourage innovation

Challenges :

Limited user freedom

Lack of transparency

High cost may restrict access

13. Consent of Informed consent in personal data collection.

Ans: Informed consent means that users clearly understand and voluntarily agree to how their personal data will be collected, used, stored and shared.

For consent to be valid, it must be:

1. Informed - users must know what data is being collected and why.
2. Voluntary → No pressure or forced agreement.
3. Specific - consent should be for a clearly defined purpose.
4. Clear & Understandable - No hidden terms or complex languages.
5. Revocable - Users should be able to withdraw consent anytime.

14. What are the ethical concerns associated with surveillance technologies? Explain with example how these technologies can be misused.

Ans: Surveillance technologies include CCTV camera, facial recognition system, location tracking and online activity monitoring. While they can improve security, they raise serious ethical concerns.

1. Violation of privacy

2. Lack of informed consent

3. Abuse of power

4. Discrimination and bias

5. Data security risks.

Example of misuse:

- Excessive govt. monitoring of citizens' online activities without proper justification
- Companies secretly tracking users' location data for profit
- Facial recognition system wrongly identifying innocent people, leading to false arrest.

15. What is Algorithmic bias? Explain it causes and suggest measures to ensure fairness in AI systems

Ans: Algorithmic bias occurs when an AI system produces unfair or discriminatory result against certain groups of people.

Causes:

- i) Biased training data (eg: AI favoring men)
- ii) Biased Algorithm design (development assumption)
- iii) feedback loops reinforcing inequalities.
- iv) Lack of diversity in development team.

Measures to ensure fairness

- i) Use diverse and representative datasets
- ii) Conduct regular audits and testing for bias
- iii) Make AI decision making transparent
- iv) Follow ethical guidelines like the ACM/IEEE code of ethics.

Q. Discuss the importance of transparency and explainability in AI. Who should be held accountable when an autonomous system causes harm?

- Ans: Importance of transparency & explainability in AI,
- Transparency ensures that AI system workings and decision making processes are open and understandable.
 - Explainability allow users, developers and regulators to understand why an AI made a particular decision.
 - Both are crucial to build trust, detect bias, prevent errors and ensure AI decisions are fair & ethical.

The Responsibility depends on context:

- Developers are accountable if poor design or lack of testing caused harm.
- Manufacturers or operators are accountable if deployment or maintenance was negligent.

Organizations deploying autonomous system hold legal and ethical responsibility for outcomes.

Clear accountability is essential to ensure victims are protected and system are improved.

18. Role of ethical leadership in IT. How can leaders foster an ethical culture within their team?

Ans: Role of ethical leadership in IT?

- i) Ethical leaders set the tone for integrity, honesty and responsible behaviour in the organization.
- ii) They ensure compliance with laws, professional codes, ethical standards.
- iii) Promote trust among employees, clients and users which improves decision making.
- iv) Encourage responsible use of technology, including data privacy, cybersecurity and fair Artificial Intelligence practices.

19. Differentiate Ethical Hacking and Cybercrime

What conditions is hacking considered ethical

Ans:

Ethical Hacking

Cybercrime

i) To identify security flaws, improve system security.

ii) Conducted with permission from the system owner.

iii) Legal and comply with laws.

iv) Outcome: strengthens security, prevents attacks.

v) Penetration testing by a security professional.

i) To steal, damage or exploit data or system.

ii) Done without consent.

iii) Illegal and punishable by law.

iv) Causes harm, privacy breaches.

v) Hack a bank account to steal money.

Hacking is considered ethical when:

- i) It is done with proper authorization from the system owner
- ii) The purpose is to improve security.
- iii) Findings are reported responsibly to help fix vulnerabilities.
- iv) It follows legal standards like ACM/IEE code of ethics.

20. Concept of responsible disclosure in cybersecurity.

What ethical guidelines should a security researcher follow when discovering a vulnerability?

Ans: Responsible disclosure is the practice where a security researcher reports a discovered vulnerability to the organization or software owner privately & securely before making it public. This allows the organization to fix the issue and protect users before attackers exploit it.

- Notify the organization first - avoid publicly sharing the vulnerability & immediately.

- Avoid exploiting the vulnerability - do not use it to steal data or cause harm.

- Be accurate & honest - provide clear details to help fix the issue.

- Follow IEEE/ACM rules.

- Coordinate disclosure timeline - work with the org to decide when public disclosure is safe.

21. What is digital divide? Discuss its social implications and suggest measures to bridge this gap.

Ans: Digital divide is the gap between people who have access to modern ICTs (internet, computers) and those who do not.

Social implications:

- Unequal education and online learning opportunities.
- Limited access to jobs and economic opportunities.
- Social exclusion from info and services.
- Reduced participation in society.

Measures to bridge the gap:

- Expand internet & mobile infrastructure.
- Provide affordable device & connectivity.
- Support govt. / NGO initiatives and local content development.

2a. What is green computing? How can green

computing professionals contribute to env. sustainability?

Ans: Green computing refers to designing, using and disposing of computers, servers and IT systems in ways that minimize env. impact. It focuses on energy efficiency, reducing e-waste and sustainable technology practises.

How computing professionals can contribute:

- Optimize programs and use low power devices.
- Use virtualization and cloud computing to reduce physical servers & energy use.
- Recycle or safely dispose of old electronics.
- Design software that uses resources efficiently.

23. What is conflict of interest? Provide example & explain.

Am: A conflict of interest occurs when a professional's personal interest clash with their duties to the employer, clients or users, potentially affecting impartiality.

Ex:

- A developer recommending a software product they have a financial stake in.
- An IT manager hiring a relative or friend for a project.

Ethical Handling:

- Disclose the conflict to relevant authorities.
- Avoid decisions that benefit personal interests over professional duties.
- Rescue themselves from decisions where impartiality is compromised.

25. What is Aristotle's Golden Mean? How does it guide

- individuals to find a balanced and virtuous course of action between extremes in ethical decision making?

Ans: Aristotle's golden mean is the idea that virtue lies between two extremes: excess and deficiency.

Ethical behaviour is about finding the balanced, moderate course of action rather than overdoing or underdoing something.

Ex:

- Courage is a virtue between recklessness (excess) and cowardice (deficiency)

- Generosity lies between wastefulness & stinginess.

□ Guidance in ethical decision making:

- Helps individuals avoid extreme behaviour.

- Encourages thoughtful, balanced choices.

- Promotes moral development by practising moderation & reason.

26. Ethical danger of Greed (Pahom's story). What lessons can computing professions learn from Pahom's downfall when making decision about profit, power & responsibility?

Ans:

Tolstoy's story 'How much Land does a man Need?' illustrates the ethical dangers of greed and excessive desire through Pahom, who constantly seeks more land. His unending ambition ultimately leads to his death, showing that excessive pursuit of wealth can destroy one's well-being and moral judgement.

Lessons for computing Professionals:

- Avoid letting profit or power override ethical responsibility.
- Consider the impact of decisions on users, society and colleagues.
- Practice moderation and integrity instead of pursuing unchecked growth or advantage.

• understand that ethical lapses in pursuit of
gain can have serious consequences; including harm,
loss of trust, or legal issues.

27. Ethical lessons from three question by Leo Tolstoy.

Ans: In three questions, Tolstoy teaches that the important time is the present moment, the most important person is the one before us, and the most important action is to do good.

The present moment matters because it is the only time we can act and make ethical choices. The person before us matters because ethical responsibility begins with those directly affected by our action. Doing good is essential because moral action should aim to benefit others and reduce harm.

Guidance for technology & professional life:

- Focus on the real impact of decision in the present.
- Consider how tech affects current users and stakeholders.

- prioritize user welfare over profit or convenience.

- Act responsibly and ethically in every situation,

28. Moral decay in Bangladesh it causes effect

and how to prevent?

Ans: Moral decay refers to the decline of

honesty, responsibility and social values in society,

Major Causes:

- Corruption and lack of accountability.

- Weak law enforcement

- Decline in Family & Moral Education.

- Economic inequality and unemployment

- Negative social and media influence,

Social and economic effect

- Loss of trust and increase in crime.

- Weakening of cultural values.

- Misuse of public resources.

- Slower economic growth.

Solution:

- Strengthen moral and ethical education
- Ensure strict enforcement of laws,
- Promote transparency & good governance.
- Encourage family & community involvement.

30. Examine how animal farm reflects the dangers of unchecked power and the moral responsibilities of leaders.

Ans: Animal farm reflects the dangers of unchecked power through the rise of pigs, especially Napoleon, who gradually becomes a Dictator. Although the rebellion began with the goal of equality & freedom, the leaders manipulate the rules and betray the original ideas for personal gain.

The film shows that when power is not limited

monitored, it leads to corruption, oppression,

inequality. The pig changes the commandments,

silence opposition and controls info to maintain

authority. This demonstrates how leaders can

misuse power when there is no accountability.

Regarding moral responsibility, leaders should:

- Serve the interests of all, not personal benefit.

- Remain transparent and honest.

- Uphold equality and justice.

- Accept accountability for their actions.

31. The Lion King: Discuss the role of mentorship and personal growth in shaping Simba's ability to confront challenges and assume leadership.

Ans: In the Lion King, mentorship & personal growth play a crucial role in shaping Simba into a responsible leader. After Mufasa's death, Simba runs away and avoids his responsibilities, however diff. mentors guide him at diff. stages of life.

Mufasa teaches Simba about courage, responsibility and the 'Circle of Life', planting the foundation of leadership. Later Rafiki reminds Simba of his true identity and encourages him to face his past. Even Timon and Pumbaa teach him about friendship and resilience, through their carefree philosophy delays his maturity.

Through these influences, Simba experiences
personal growth. He learns to confront his
fears, accept responsibility, and understand
that leadership requires courage and
accountability.

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